

**What makes an
asset the
dominant
intermediary?**



**Centre for Blockchain
Technologies**

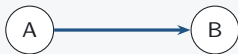
The Making of Dominant Vehicle Currencies

Evidence from DeFi
Jiahua Xu

Transaction-level route structures

Direct swap

hash h_D ; component 0



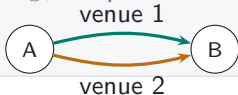
Sequential vehicle route

hash h_V ; component 0



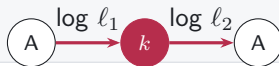
Parallel split

hash h_S ; component 0



Round trip

hash h_R ; component 0

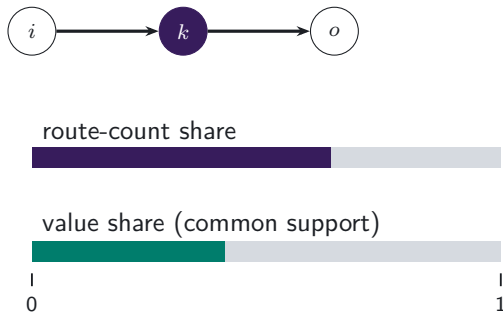


The hash groups swap events. Token connectivity assigns components, and log index orders legs within the transaction.

Measuring vehicle dominance

MEASUREMENT

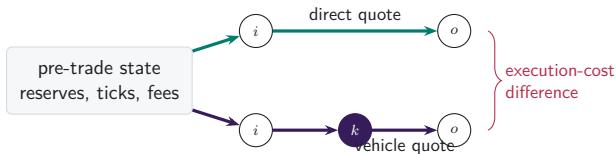
- Dominance is the share of indirect trade intermediated by asset k .
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.



The blockchain reveals chosen routes and contemporaneous alternatives

RESEARCH DESIGN

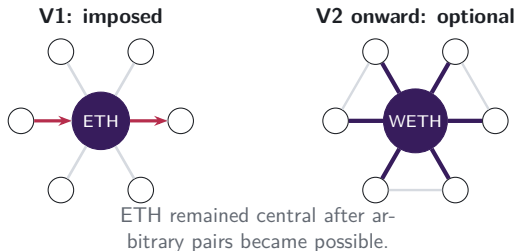
- FX data typically reveal executed trades.
- On-chain reserves and fees also let us price feasible alternatives.
- The comparison holds market conditions fixed at execution.



Protocol design first imposed ETH intermediation

INSTITUTIONAL FACT

- Uniswap V1 created one exchange contract per token, paired with ETH.
- Token-to-token swaps therefore moved through ETH by construction.
- V2 allowed arbitrary pairs; the ETH-centred pairing network persisted.



The panel follows the market across six years

DATA

2,332

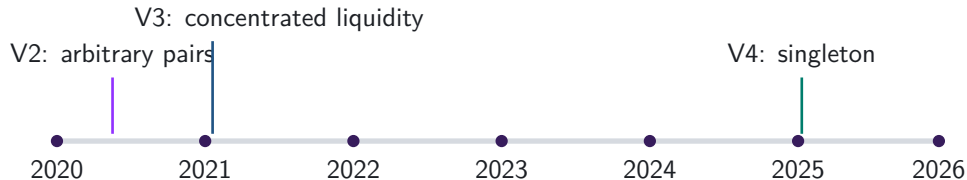
calendar days

471.6m

swap legs in reconstructed
routes

8

DEX deployments



Routes are reconstructed transaction by transaction, then aggregated to daily and weekly observations.

Four measures capture use, scale, and network position

1. Vehicle share

Fraction of route units intermediated by asset k .

2. Excess use

Intermediary share relative to endpoint demand.

3. Network position

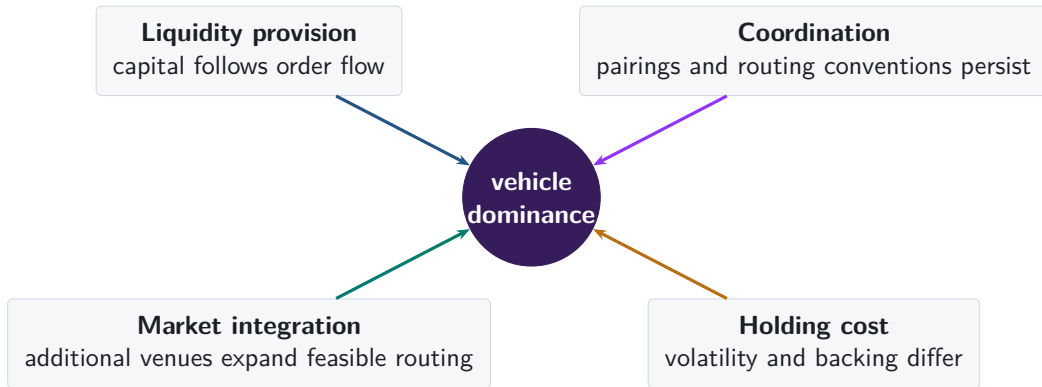
Degree, strength, and betweenness on the trading graph.

4. Execution advantage

Direct-route output relative to a feasible vehicle route at the same pre-trade state.

Together, the measures distinguish use, scale, network position, and execution cost.

Four forces can sustain vehicle dominance



The empirical design separates formation, rotation, and persistence

FORMATION

Architecture changes

Imposed intermediation, arbitrary pairing, concentrated liquidity, and singleton settlement.

ROTATION

Panel variation

Count share, value share, excess use, network position, and route complexity over time.

PERSISTENCE

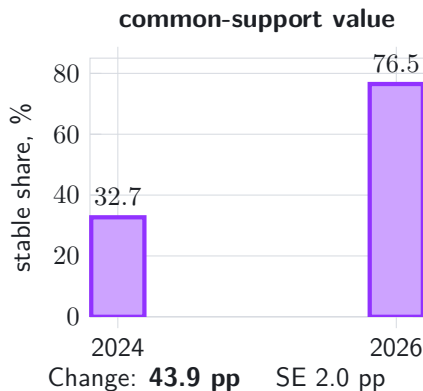
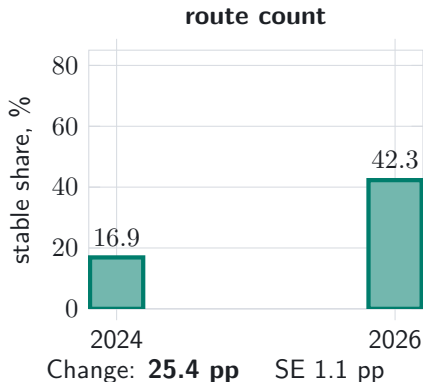
Contemporaneous alternatives

An incumbent retains volume despite a cheaper feasible route.



Stablecoins gained vehicle share by count and by value

DESCRIPTIVE RESULT

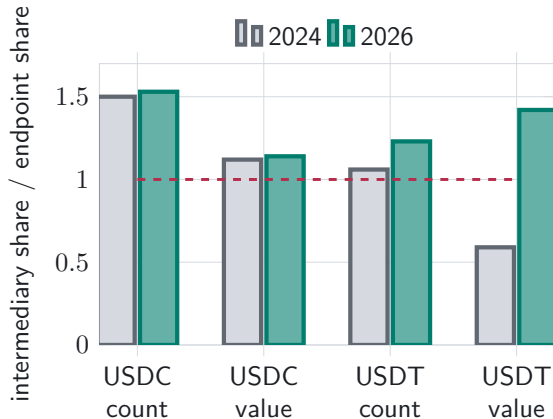


Observed two-leg routes with comparable dollar values; daily shares are equally weighted within endpoint-year.

USDT drives the shift toward stablecoin intermediation

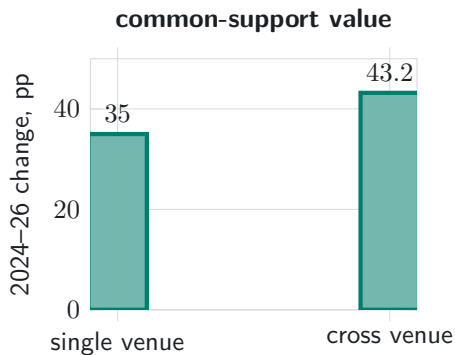
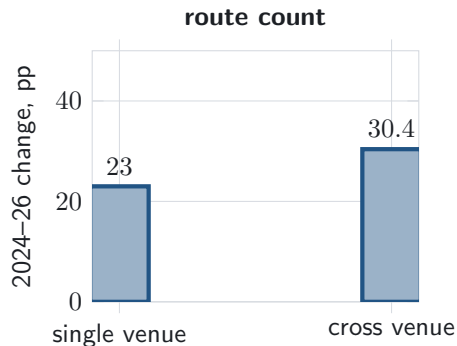
TOKEN DECOMPOSITION

- USDC was already used more as an intermediary than its endpoint share implies.
- USDT shifted from route endpoints toward intermediation.
- USDC and USDT account for 92.1% of the stablecoin count-share increase.



Rotation is present within venues and stronger across venues

MARKET INTEGRATION



The increase is **7.5 pp** larger for cross-venue routes by count (SE 1.3) and **8.2 pp** larger by value (SE 2.3).

Dominance reflects coordination, demand, and market design

1. **A protocol can create a vehicle role.**

V1 imposed ETH intermediation; the ETH-centred pairing network persisted after arbitrary pairs became possible.

2. **The dominant vehicle can rotate.**

Stablecoins gained 25.4 percentage points by route count and 43.9 points by common-support value from 2024 to 2026.

3. **USDT drives the transition.**

Its intermediary share rose above its endpoint share, and the rotation is stronger in cross-venue routing.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit



one route unit
two swap legs

- Each linked $i \rightarrow k \rightarrow o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k .

A2. Backing regimes vary over time

Native platform asset

ETH, WETH; volatile settlement asset and protocol-native demand.

Fiat-reserve stablecoin

USDC, USDT; reserve and redemption design.

On-chain collateralised

DAI, LUSD, crvUSD; collateral, liquidation, and backing changes.

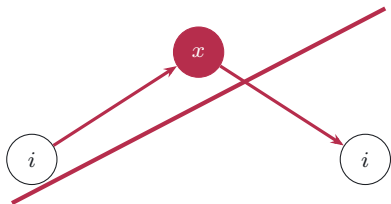
Tokenised external asset

WBTC, tokenised gold; wrapped claim on an external asset.

Heterogeneity tests use each asset's contemporaneous backing regime.

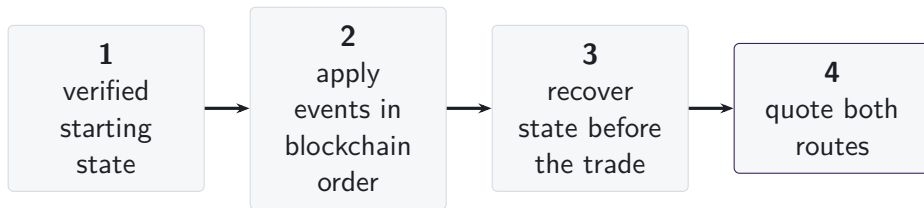
A3. Sample restrictions preserve value measurement

- Round trips are excluded from intermediation measures.
- Dollar values require consistent prices and verified token identities.
- We retain routes whose two legs imply dollar values within the pre-specified tolerance.



round trip: excluded from
value-weighted vehicle measures

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. AMM designs require different state variables

Constant product

Reserves, fee, invariant. **Concentrated**

liquidity

Active liquidity, price, tick, and initialised-tick changes. **StableSwap**

families

Balances, amplification, rates, and oracle state.

Estimates include protocol families whose balances, fees, and pricing state can be independently validated.

Weighted pools

Vault balances, weights, scaling, and rate providers. **V4 singleton**

Pool key, tick state, and fee for standard pools without custom hooks.

A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- The paired legs report equal ETH quantities in 87.4% of identified forced routes.
- This two-leg signature identifies protocol-imposed intermediation.



217,003
forced routes

A7. Daily and weekly frequencies answer different questions

- | | |
|----------------------------------|---|
| 1. Primary panel | day observations; date fixed effects |
| 2. Calendar sensitivity | day observations; week and weekday fixed effects |
| 3. Aggregation robustness | complete weeks; week fixed effects |
| 4. Transition series | exact 1-, 7-, 30-, and 120-day links; HAC inference |

Weekly ratios are formed from weekly sums; results are reported for weeks ending on each day of the week.

A8. Count and value define separate dominance margins

Count-weighted share

$$S_{k,t}^N = \frac{N_{k,t}^I}{\sum_j N_{j,t}^I}$$

Each reconstructed route carries one unit of weight.

Value-weighted share

$$S_{k,t}^V = \frac{\text{Vol}_{k,t}^I}{\sum_j \text{Vol}_{j,t}^I}$$

Each route carries reliably priced dollar value on common support.

Differences between count and value shares may reflect trade size, route structure, or the mix of intermediary assets.

A9. USDT gains intermediary use beyond endpoint demand

Route-count share

- Intermediary-minus-endpoint share rises by **2.39 pp**.
- HAC s.e.: 0.89 pp.

Dollar-value share (common support)

- Gap moves from -7.13 **pp** to $+8.14$ **pp**.
- Change: 15.27 pp; HAC s.e.: 1.50 pp.

USDT's intermediary share rises above its endpoint share.

A10. Capital, inventory, and depth are distinct quantities

Deposited capital

Funds committed by liquidity providers, established from independently valued holdings or audited claims. **Inventory**

Exact token amounts held by the pool.

Capital enters only where holdings and ownership are independently validated; execution depth comes from the pool's pricing function.

Executable depth

Amount that can be traded for a given price impact under current pool state.

Quote quality

Execution price for trade size q along feasible routes.

A11. Integration changes both reach and route choice

Available trading opportunities

- active venues
- feasible paths
- available intermediary assets

Route choice conditional on availability

- direct versus intermediated
- single- versus cross-venue
- sequential versus split routing

We test whether stablecoin intermediation rises more when additional venues and routes are available, conditional on endpoint pair and date.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Millionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage